DataCORE

Advanced Pandas, Publication-Quality Figures & Tables

Afternoon: DataCORE

1. **Advanced Pandas — GroupBy, Pivot Tables, Merging** (1:00–2:00)
2. **Publication-Quality Figures** (2:00–2:45)
3. **Scientific Storytelling — Table Captions** (2:45–3:00)

*This morning you met your data. This afternoon you **slice it by groups**, build **publication-ready tables and figures**, and learn to **caption** them like a journal article.*

1:00 – 2:00 · Advanced Pandas

From one variable to **many**

- So far: one variable at a time (a mean, a histogram).
- Real analysis asks **comparative** questions: *does the pattern differ **by group**?*
- Three workhorse tools do almost all of it:

groupby — summaries within each group

pivot_table — publication-style tables

merge — combine two datasets on a key

Our data: the same wage survey — `education`, `gender`, `region`, `experience_years`, `hourly_wage`.

Load the data

Python R

```
wages = pd.read_csv("wages.csv")
print(wages.head())
```

```
     education  gender    region  experience_years  hourly_wage
0     Bachelor    Male     Rural                10        28.54
1    Associate    Male     Urban                11        28.03
2     Graduate  Female  Suburban                 4        31.56
3     Bachelor  Female     Urban                16        24.90
4  High School    Male     Urban                16        27.83
```

```
print("N =", len(wages))
```

```
N = 220
```

One row per worker. We'll summarise **across** workers, grouped by their characteristics.

GroupBy — multi-level summaries

Python R

```python
print(wages.groupby(["education", "gender"])["hourly_wage"]    # group by TWO keys
          .agg(["mean", "std", "count"]).round(2))
```

```
                    mean   std  count
education  gender
Associate  Female  22.80  3.74     19
           Male    25.58  5.58     26
Bachelor   Female  28.05  4.20     40
           Male    30.42  3.57     40
Graduate   Female  33.38  4.33     17
           Male    36.45  2.81     10
High School Female 20.64  4.19     35
           Male    24.09  4.01     33
```

Grouping by **two** keys gives one row per *combination*. `agg([...])` computes several statistics at once — mean, spread, and sample size together.

Pivot table — a publication-style layout

Python R

```
print(pd.pivot_table(wages, values="hourly_wage",
                     index="education", columns="gender",
                     aggfunc="mean").round(2))
```

```
gender        Female   Male
education
Associate      22.80  25.58
Bachelor       28.05  30.42
Graduate       33.38  36.45
High School    20.64  24.09
```

A pivot turns **groups into rows and columns** — the exact shape of a results table you'd drop into a paper or slide.

Merge — combine two datasets on a key

Python R

```python
region_info = pd.read_csv("region_info.csv")          # cost of living by region
merged = pd.merge(wages, region_info, on="region")    # join on the shared key
print(merged.head())
```

```
     education  gender    region  experience_years  hourly_wage  cost_of_living_index  median_rent
0     Bachelor    Male     Rural                10        28.54                    88          950
1    Associate    Male     Urban                11        28.03                   121         1980
2     Graduate  Female  Suburban                 4        31.56                   104         1450
3     Bachelor  Female     Urban                16        24.90                   121         1980
4  High School    Male     Urban                16        27.83                   121         1980
```

Merging matches rows by a **shared key** (here `region`) — the same way you'd join county health data to county income data on a **FIPS code**.

Team Task

🏁 Which groupby matters for **your** question? — ~12 min

1. Pick the **grouping variable** most relevant to your research question.
2. Run `df.groupby('group')['outcome'].agg(['mean','std','count'])` (Python) **or** the `group_by() |> summarise()` (R).
3. **Stretch:** group by **two** variables, or build a `pivot_table`.

Then answer in one sentence:

*"Grouped by **[variable]**, the **[outcome]** was highest for **[group]** (mean = __) and lowest for **[group]** (mean = __)."*

What does that tell you about your question?

12:00

2:00 – 2:45 · Publication-Quality Figures

One rule

If you would be embarrassed to show this chart in a conference presentation, do not put it in your slides.

Readable
Clear axis labels with units, a title, legible font sizes.

Honest
Show uncertainty (error bars); don't truncate axes to exaggerate.

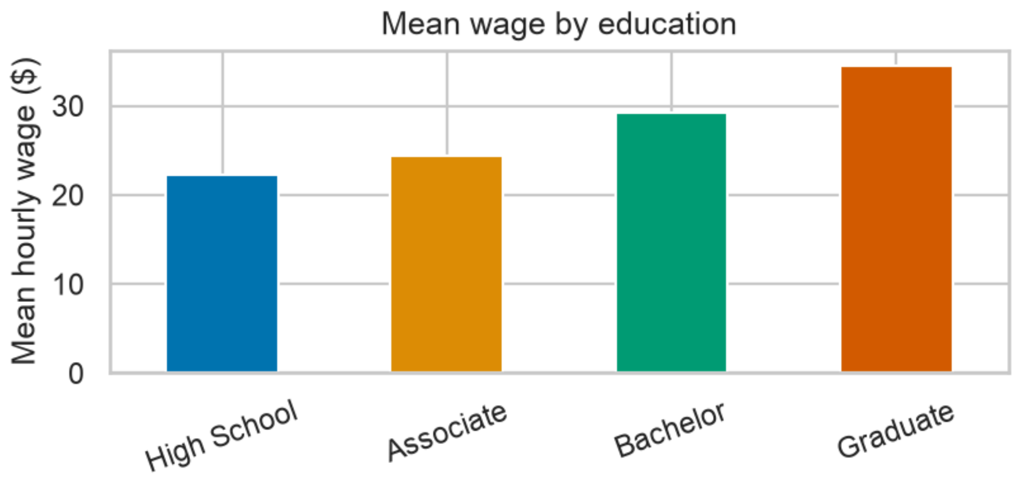
Accessible
Colorblind-safe palettes; don't rely on red/green alone.

We'll upgrade a plain chart step by step: **palette** → **error bars** → **annotation** → **export at 300 DPI.**

Colorblind-safe palettes

Python (seaborn) | R (ggplot2)

```
order = wages.groupby("education")["hourly_wage"].mean().sort_values()
colors = sns.color_palette("colorblind", len(order))      # colorblind-safe
order.plot(kind="bar", color=colors)
plt.ylabel("Mean hourly wage ($)"); plt.xlabel("")
plt.title("Mean wage by education")
plt.xticks(rotation=20); plt.tight_layout(); plt.show()
```

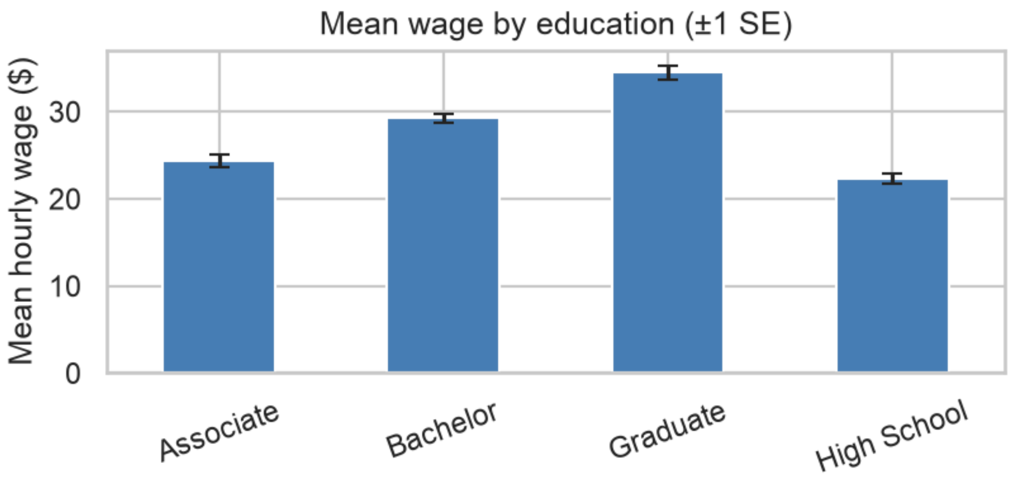


`seaborn`'s **"colorblind"** palette and ggplot's **viridis** are designed to stay distinguishable for color-vision deficiencies.

Error bars — show the uncertainty

Python R

```
g = wages.groupby("education")["hourly_wage"]
means = g.mean(); ses = g.std() / np.sqrt(g.count())       # standard error
means.plot(kind="bar", yerr=ses, capsize=4, color="#4a7fb5")
plt.ylabel("Mean hourly wage ($)"); plt.xlabel("")
plt.title("Mean wage by education (±1 SE)")
plt.xticks(rotation=20); plt.tight_layout(); plt.show()
```



A bar without error bars hides how **certain** the average is. **mean ± 1 SE** is the standard add-on.

Annotate the point that matters

Python   R

```
x = wages["experience_years"]; y = wages["hourly_wage"]
plt.scatter(x, y, alpha=0.5, color="#4a7fb5")
i = y.idxmax()                                              # highest earner
plt.annotate("highest earner", (x[i], y[i]), xytext=(x[i]-13, y[i]+1),
             arrowprops=dict(arrowstyle="->", color="#D55E00"))
plt.xlabel("Years of experience"); plt.ylabel("Hourly wage ($)")
plt.tight_layout(); plt.show()
```

One well-placed label guides the eye to your **key data point** — far better than making the viewer hunt for it.

Export at 300 DPI for slides & print

Python | R

```python
plt.savefig("fig1_wage_by_education.png",
            dpi=300, bbox_inches="tight")      # crisp; no clipped labels
```

Give every figure a **descriptive file name** (`fig1_wage_by_education.png`, not `Untitled.png`) and a **caption**. `bbox_inches="tight"` stops axis labels from being cut off.

Team Task

🏁 Your 2–3 best charts — ~12 min

1. Choose the **2–3 charts** that best answer your research question.
2. Upgrade each: **colorblind-safe palette**, clear labels with **units**, a title.
3. Add **error bars** or **annotate** the key point where it helps.
4. **Save** each at **300 DPI** with a descriptive file name.

Each chart needs:

- A descriptive **file name**
- A one-line **caption**
- A reason it's in the deck — *what question does it answer?*

12:00

2:45 – 3:00 · Scientific Storytelling

A table is useless without a caption

Python | R

```
print(wages.groupby("education")["hourly_wage"]
          .agg(["mean", "median", "std", "min", "max"]).round(2))
```

```
              mean  median   std    min    max
education
Associate    24.41   23.53  5.04  12.13  34.17
Bachelor     29.24   29.10  4.05  19.06  37.72
Graduate     34.51   34.98  4.07  26.16  42.03
High School  22.32   22.22  4.43  10.96  31.28
```

A reader who sees only this table should still know **what it shows, where it came from, and one thing to notice.** That's the caption's job.

APA table format

> **Table 1.** *Hourly wages by education level (N = 220 workers).* **Note.** Data from the 2023 wage survey; values are in U.S. dollars per hour. Mean hourly wage rises with education, from \$22 (high school) to \$35 (graduate).

The shape:
*"Table 1. **[Descriptive title]**. Note. **[Source and any key definitions]**."*

Why it matters:
APA table format is the **standard** in the social and health sciences — titles above the table, notes below.

Write your table's caption — now

Write a caption for **your** summary-statistics table:

> *"Table 1. **[descriptive title]**. Note. **[source, sample, definitions]**; **[one notable pattern]**."*

Read it to a partner who hasn't seen your data.

Checklist:

- Descriptive **title** (not just "Table 1")?
- **Sample / N** stated?
- **Source & year**?
- Units / key **definitions**?
- **One pattern** worth noticing?

10:00